

Dan Kim

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SUMMARY

Graduate Research Student at Georgia Tech with expertise in Python, data pipeline development, and machine learning. Developed automated ETL pipelines for high-throughput classification and interpretable ML models achieving ~0.70 AUC. Skilled in data preparation, statistical analysis, and version control, with experience collaborating in cross-functional research teams. Seeking an AI & ML grad hires to apply data science and AI/ML skills.

SKILLS

- **Programming | Frameworks:** Python, MATLAB, Linux, Git, ROS2
- **AI/ML:** PyTorch, Scikit-learn
- **Data Analysis & Statistics:** SQL, Statistics, Data Analysis

EDUCATION

Georgia Institute of Technology, GA

Master of Science, Robotics (Artificial Intelligence)

• **GPA:** 3.79/4.00

• **Coursework:** Machine Learning, Machine Learning with Neural Data, Computer Vision, Human Robot Interaction

Aug 2024 - May 2026

Atlanta

University of Illinois at Urbana Champaign, IL

Bachelor of Science, Mechanical Engineering

• **GPA:** 3.78/4.00

• **Achievements:** Graduation with Honors

Jan 2017 - Dec 2021

Urbana

WORK EXPERIENCE

Georgia Institute of Technology

Graduate Research Student

Atlanta · **Nov 2024 - June 2026**

- Engineered a Python-based extract, transform, and load (ETL) pipeline to convert raw, unstructured JSON logs into structured, analysis-ready datasets, optimizing classification throughput.
- Developed interpretable machine learning models using scikit-learn (regression and decision trees) to identify patterns in non-stationary time-series data, achieving ~0.70 AUC and surfacing potential biomarkers via game-theoretic interpretability.
- Conducted statistical analysis with MATLAB and Python on physiological datasets to pinpoint brain regions associated with heart-rate modulation.

Graduate Teaching Assistant

Atlanta · **May 2025 - May 2026**

- Mentored mathematical models and utilized numerical approximation methods (root-finding, curve fitting) to solve complex engineering problems and apply to experiments, achieving high precision in data output
- Led over 120 students in regression analysis and curve fitting to identify data trends, generate confidence intervals, and validate experimental results statistically.

Korea Institute of Industrial Technology (gov)

Robotics Automation Research Intern

South Korea · **Nov 2023 - Jun 2024**

- Applied machine learning to automate image classification for a robotic station, projected to reduce workflow time by 30% upon implementation.
- Reviewed custom Python software for hardware-software integration, optimizing system performance and data throughput
- Executed rigorous QA protocols across 60+ system inputs/outputs to ensure the integrity and reliability of automated data streams

Companoid Labs

Product Insight Analyst (UX)

South Korea · **Apr 2022 - Jun 2023**

- Developed a persuasive onboarding process for digital therapeutics (DTx), enhancing user attitude by 15%
- Presented behavioral research findings at Korea's HCI conference, focusing on the intersection of digital therapeutics and user engagement
- Developed standardized research and validation frameworks to test hypotheses, leading a team of interns
- Synthesized qualitative and quantitative user data from interviews and surveys to identify friction points and inform products

Republic of Korea Army

Information Operations Personnel

South Korea · **Jul 2017 - Apr 2019**

- Secured sensitive communications via encryption protocols to maintain battalion-wide data integrity
- Awarded "Best Decryption Personnel" in a battalion-wide competition, demonstrating proficiency in pattern recognition

PROJECTS

A.C.E.S – Agentic Control and Execution System for automated trading

May 2026

- Implemented and upgraded a static one-time multi-agent framework to a 24/7 control system in Python, coordinating specialized LLM agents via LangChain and LangGraph for real-time data processing and automated state-based command execution
- Designed a unified data model systematically classifying system entities (account metrics, active positions) and established safety boundaries such as dynamic stop/loss limit based on volatility calculations to enforce data quality and system reliability real-time.
- Handled real-time AI model inference and structured JSON result consumption, exposing model evaluations, execution logs, and live data via an interactive Python-based dashboard

Deep Learning for Recognition & Segmentation

Jan 2025 - May 2025

- Built and trained convolutional neural networks in Python using PyTorch for large-scale scene recognition, achieving ~70% classification accuracy
- Engineered semantic segmentation models to perform pixel-level classification, utilizing architectures like ResNet for feature extraction

Electroencephalogram (EEG) Motor Imagery signal classification

May 2025 - Jul 2025

- Engineered CNN and Temporal Convolutional Network (TCN) architectures to classify high-dimensional, 4-class motor imagery signals from the BCI Competition IV-2a dataset.
- Achieved 70% classification accuracy on raw time-series EEG data by optimizing neural decoder architectures for nonstationary signal patterns

Autonomous Driving Maze Robot

Aug 2024 - Dec 2024

- Developed a computer vision pipeline utilizing image preprocessing and ML classification (SVM) to identify road signs for autonomous robotic navigation.
- Optimized navigation parameters within the ROS2 Nav2 framework, utilizing YAML-based configuration to tune real time sensor-to-actuator latency